Supervised Learning (Labeled Data, Predict Outcomes)

- **How it works:**

Uses labeled input-output pairs to train models to map inputs to known outputs.

- **Goal:** Predict outcomes for new, unseen data.
- **Examples:** Image classification, spam filtering, demand forecasting.
- **Types:** Classification (predicting categories) and Regression (predicting numerical values).

Unsupervised Learning (Unlabeled Data, Find Patterns)

- **How it works:** Analyzes unlabeled data to discover hidden patterns, structures, or features.
- **Goal:** Group or compress data, finding relationships within data points.
- **Examples:** Customer segmentation, recommendation engines, anomaly detection.
- **Types:** Clustering (grouping) and Dimensionality Reduction (reducing complexity).

Reinforcement Learning (Trial-and-Error, Reward-Based)

- **How it works:** An agent interacts with a dynamic environment, taking actions to achieve a goal and receiving feedback (rewards or penalties).
- **Goal:** Learn the best strategy (policy) to maximize cumulative rewards.
- **Examples:** Game AI (e.g., AlphaGo), robotics, self-driving cars.